

Documentation on XAI

****XAI- reference papers;**

.)A Survey on Explainable Artificial Intelligence (XAI): Towards Medical XAI;

.Explainable Artificial Intelligence Approaches: A Survey;

=> talks about the different features that are used for explaining the predictions done by an ai model and also to understand the accuracy of the predicitions as well as explanations using both mathematical and practical ways;

topics useful for our project:

.Clustering logic implemented by the kubernetes Ai model can be validated by the user using techniques like LIME ,RULE_SETS, TCAV,t-SNE

.Generation of YAML files for the kubernetes logic;

.SHAP techniques will be useful for the decision analysis

****)**Metrics required for XAI in terms of kubernetes development

.Explanation Fidelity

.Explanation Coverage

.Counterfactual Quality

.Deployment Success Rate

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Scope of prometheus and grafana:

Prometheus in Project

Prometheus acts as the **data collector**. It continuously gathers Kubernetes metrics such as:

- CPU usage
- Memory usage
- Pod health
- Replica count
- Traffic/load

It can also collect your **custom AI/XAI metrics** like:

- Fidelity Score
- Confidence Score
- Coverage Score
- Deployment Success Rate

These metrics become the input for your AI and are also stored for monitoring.
